

Let: x = amount borrowed at 3%.

y = amount borrowed at 4%.

z = amount borrowed at 5%.

Then, we get 3 equations: $x + y + z = 225000$

$$0.03x + 0.04y + 0.05z = 9900$$

$$y = 2.5x$$

Substitute $y = 2.5x$ into the first equation:

$$x + 2.5x + z = 225000$$

$$3.5x + z = 225000$$

$$z = 225000 - 3.5x$$

Sub. into the interest equation: $0.03x + 0.04(2.5x) + 0.05(225000 - 3.5x) = 9900$

$$0.03x + 0.04(2.5x) + 0.05(225000 - 3.5x) = 9900$$

$$-0.045x + 11250 = 9900$$

$$-0.045x = -1350$$

$$x = 30,000$$

Find y and z : $y = 2.5(30000) = 75,000$

$$z = 225000 - 3.5(30000) = 120,000$$

Answer: \$30,000 at 3%, \$75,000 at 4%, \$120,000 at 5%